

COURSE TITLE : FUNDAMENTALS OF CHEMICAL ENGINEERING
COURSE CODE : 2071
COURSE CATEGORY : B
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Introduction to Chemical Engineering.	15
2	Unit Operations	15
3	Unit Processes and Basic Concept of Chemical Processes	15
4	Process Instrumentation and Safety	15
TOTAL		60

COURSE OUTCOMES :

Sl.No	Sub	Student will be able to
1	1	Know the historical background of Chemical Engg and familiarize the fundamentals
	2	Understand the types and nature of industries.
	3	Gain the knowledge of Unit operations in chemical industries.
2	1	Know the basic concepts of Unit processes in chemical plants.
	2	Understand measurements and instrumentation used in Chemical processes .
	3	Know safety precautions in Chemical plants

SPECIFIC OUTCOMES

MODULE- I INTRODUCTION TO CHEMICAL ENGINEERING

1.1.0 Understand the historical background and scope of Chemical Engineering.

- 1.1.1 Differentiate types of industries, Nature of industries.
- 1.1.2 List the major chemical industries in India
- 1.1.3 Identify the location of the major chemical industries in Kerala
- 1.1.4 List the product and raw materials of ten major industries
- 1.1.5 Explain the importance of chemical industry

1.2.0 Know the duties of a Chemical Engineer

- 1.2.1 Compare the duties and responsibilities of a chemist and a chemical engineer
- 1.2.2 List the duties & tasks of a Chemical engineering technician in Industry
- 1.2.3 Explain the principle of development of a chemical plant
- 1.2.4 Explain the principles of economy, chemical feasibility in chemical industries

1.3.0 Understand basic chemical calculations

- 1.3.1 Explain various system of units and its conversions
- 1.3.2 Solve simple problems related to units and its conversions.
- 1.3.4 Define concept of mole, molecular weight, equivalent weight, normality, molarity, molality.
- 1.3.5 Solve elementary problems related to the above definitions.

1.4.0 Understand Gas laws in chemical processes.

- 1.4.1 State Gas Laws such as Ideal gas law, Dalton's Law, Amagat's Law
- 1.4.2 Solve simple problems related to above Gas laws.

MODULE- II UNIT OPERATION

2.1.0 Understand various Unit operations in Chemical Engineering

- 2.1.1 Define and explain the terms in Mechanical operations such as Size reduction, size separation, filtration, sedimentation, centrifugation and mixing with symbols.
- 2.1.2 Explain the important properties of fluid
- 2.1.3 List the methods of handling and storage of liquids.

2.2.0 Understand different mass transfer operations.

- 2.2.1 Explain the important mass transfer operations like Gas absorption, desorption, Extraction and leaching.
- 2.2.2 Describe the heat and mass transfer operations like Distillation, Drying & Crystallisation.

2.3.0 Know the different methods of Heat transfer

- 2.3.1 Define heat, temperature & sensible heat.
- 2.3.2 Explain the modes of heat transfer
- 2.3.3 Explain heat exchanging equipments.

MODULE- III UNIT PROCESSES AND BASIC CONCEPT OF CHEMICAL PROCESSES

3.1.0 Understand important unit processes.

- 3.1.1 Define unit processes with examples like sulphonation, polymerisation, oxidation, reduction, hydrogenation, hydration, saponification, alkylation, esterification, nitration, chlorination, calcination, and cracking/ pyrolysis.

3.2.0 Know basic concept of chemical process.

- 3.2.1 Define conversion, yield, and reaction efficiency.
- 3.2.2 Explain flow diagram, flow sheets and block diagram.

MODULE- IV PROCESS INSTRUMENTATION AND SAFETY

4.1.0 Know the measuring systems.

- 4.1.1 List temperature scales, measurement of temperatures using bimetallic thermometer, mercury expansion thermometer, gas/vapour filled thermometer.
- 4.1.2 List the pressure scales and units
- 4.1.3 Explain the measurement of pressure using manometers, Bourdon tube.
- 4.1.3 Explain level measurement using direct methods like bob-tape, float-tape, Sight glass etc
- 4.1.4 Describe flow measurement using Rotameter.
- 4.1.5 Explain viscosity measurement using Redwood viscometer.

4.2.0 Understand the Personal Protective Equipments.

- 4.2.1 Explain the requirement of Personal Protective equipments.
- 4.2.2 Classify the personal protective equipments used in industry.
- 4.2.3 Select suitable personal protective equipment for the following:
 - a) PPE against electric shock
 - b) PPE for eyes
 - c) PPE for face and eyes
 - d) PPE for arms, hands, and fingers
 - e) PPE for foot and leg

- f) PPE for head
- g) PPE for ear
- h) PPE for respiration system
- 4.2.4 Explain mechanical filter respirator
- 4.2.5 Describe chemical cartridge respirators
- 4.2.6 Explain air line respirator
- 4.2.7 Explain self contained breathing equipment
- 4.2.8 Explain the Safety belt for industrial operation

COURSE CONTENT

MODULE- I INTRODUCTION TO CHEMICAL ENGINEERING

Types of industries- Nature of industries.-major chemical industries in India -location of the major chemical industries in Kerala-product and raw materials of ten major industries-importance of chemical industry.

The duties of a Chemical Engineer-duties and responsibilities of a chemist and a chemical engineer-duties & tasks of a Chemical engineering technician in Industry -development of a chemical plant- chemical feasibility in chemical industries and the purification process

Basic chemical calculations-various system of units and its conversions

simple problems related to units and its conversions.-concept of mole, molecular weight, - equivalent weight, normality, molarity, molality.-Ideal gas law, Dalton's Law, Amagat's Law

MODULE- II UNIT OPERATIONS IN CHEMICAL ENGINEERING

Size reduction, size separation, filtration, sedimentation, centrifugation and mixing with symbols.

-properties of fluid -handling and storage of liquids.

different mass transfer operations .-Gas absorption, desorption, Extraction and leaching.

-Distillation, Drying & Crystallisation.

different methods of Heat transfer-heat, temperature & sensible heat.-the modes of heat transfer

- heat exchanging equipments.

MODULE- III UNIT PROCESSES AND BASIC CONCEPT OF CHEMICAL PROCESSES

Important unit processes. -sulphonation, polymerisation, oxidation, reduction, reduction, hydrogenation, hydration, saponification, alkylation, esterification, nitration, chlorination, calcination, and cracking/ pyrolysis.

Basic concept of chemical process. -conversion, yield, and reaction efficiency -flow diagram, flow sheets and block diagram

MODULE- IV PROCESS INSTRUMENTATION AND SAFETY

Measuring systems-temperature scales, measurement of temperatures using bimetallic thermometer, mercury expansion thermometer, gas/vapour filled thermometer-pressure scales and units

-measurement of pressure using manometers, Bourden tube. - level measurement using direct methods like bob-tape, float-tape, Sight glass etc-flow measurement using Rotameter- viscosity measurement using Redwood viscometer.

PERSONAL PROTECTIVE EQUIPMENTS.

- a) PPE against electric shock
- b) PPE for eyes
- c) PPE for face and eyes
- d) PPE for arms, hands, and fingers
- e) PPE for foot and leg
- f) PPE for head
- g) PPE for ear
- h) PPE for respiration system
mechanical filter respirators-

REFERENCE

- | | | | | |
|------------------------|---|---|---|-------------|
| Badger & Banchemo | - | Introduction to Chemical Eng. | - | McGraw Hill |
| Butt & Vora | - | Stoichiometry | | |
| Dryeden | - | Chemical Technology | | |
| R K Jain, Sunil S. Rao | - | Industrial Safety in Health and Environmental
Management Systems | | |